C&G

C&G, in order to increase its revenues, has decided to launch a new product. The Marketing Manager, Dr. Rossi, has submitted two alternatives to the General Manager: the product Esse-iper, which would represent an incremental modification of a product currently sold, and the product Ti, which would be a completely new product for C&G.

The study prepared by Dr. Rossi provides sales estimates for the two products over the six years following their introduction (see table below).

Year	1	2	3	4	5	6
Esse-iper (thousands unit)	350	310	250	200	100	90
Ti (thousands unit)	100	200	300	300	400	450

The expected selling prices are €40 for Esse-iper and €55 for Ti, while the unit variable costs are €25 and €30, respectively.

- The detailed design and engineering of the two products would cost €6 million and €10 million, respectively; both investments would be undertaken at year 0. Given the nature of the products, these costs are depreciable over 4 years (starting from year 1).
- The company incurs annual marketing costs of €500,000.
- The company decides to take out a bank loan of €1 million (starting from year 0) to finance another product called "Doppia W."
- The launch of the products would require advertising expenditures of €2 million (Esse-iper) and €3 million (Ti), to be incurred in year 1; these expenses are depreciable over 2 years (starting from year 2).

Knowing that:

Dr. Rossi worked 20 man-days on the preparation of the proposal, at a cost of €500 per day.

The cost of equity for C&G is 25%, and the corporate tax rate is 30%.

Determine which of the two investments is more convenient for C&G using the Net Present Value (NPV) criterion.

Calculate the payback period for both investments.

Describe qualitatively what would happen if the company's cost of equity were to increase.